Lahore University of Management Sciences
Syed Babar Ali School of Science and Engineering
Fall 2024
Final Exam – Part II – AI-500

Weightage: 55%

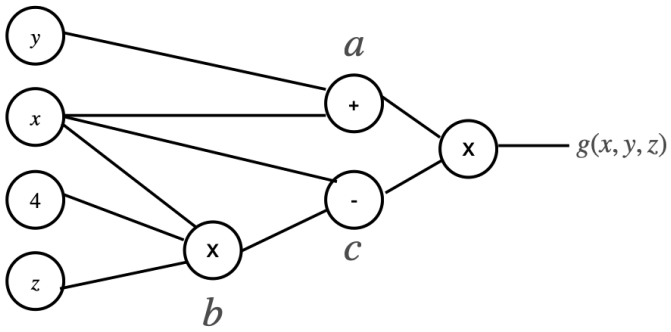
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Note: Attempt all the questions (total 40 points)
Time allowed: 1 hour 30 minutes

Problem 1. For the given equation

$$g(x, y, z) = (x - 4xz)(x + y) \tag{1}$$

(a) (**2 pts**) Construct the computational graph.



(b) (**2 pts**) For the values $x = 1$, $y = 3$ and $z = -3$, compute the forward pass. Clearly define all the intermediate values both on the graph as well as in your working. (Note: direct solution will not be awarded any points).

$a = (x + y) = 1 + 3 = 4$
$b = 4xz = 4 \times 1 \times (-3) = -12$
$c = x - b = 1 - (-12) = 13$
$g = c \times a = 13 \times 4 = 52$

(c) (**4 pts**) Using **chain rule** compute the $\partial g/\partial x$, $\partial g/\partial y$, and $\partial g/\partial z$. Clearly mention all the intermediate steps both on the graph as well as in your working (you may need to draw multiple copies of graph to clearly show different stages of backward pass). (Note: direct solution will not be awarded any points).

$$\frac{\partial g}{\partial y} = \frac{\partial g}{\partial a}\frac{\partial a}{\partial y} = c = x - 4xz - 1 - 4(-3) = 13 \tag{2}$$

$$\frac{\partial g}{\partial z} = \frac{\partial g}{\partial c}\frac{\partial c}{\partial b}\frac{\partial b}{\partial z} = a(-1)4x = -4x(x + y) = -4(1 + 3) = -16 \tag{3}$$

$$\frac{\partial g}{\partial z} = \frac{\partial g}{\partial a}\frac{\partial a}{\partial x} + \frac{\partial g}{\partial c}\frac{\partial c}{\partial x} + \frac{\partial g}{\partial c}\frac{\partial c}{\partial b}\frac{\partial b}{\partial x} \tag{4}$$

$$= c + a + a(-1)4z = x - b + (x + y) + (x + y)(-1)4z \tag{5}$$

$$= x - 4xz + (x + y) + (x + y)(-1)4z = x - 4xz + x + y - 4xz - 4yz \tag{6}$$

$$= 2x + y - 8xz - 4yz = 2 + 3 + 24 + 36 = 65 \tag{7}$$

Problem 2. We are given three points as follows:

x	1	2	3
y	2	6	12

We want to fit a quadratic model of the form:

$$y = ax^2 + bx + c \tag{8}$$

We want to find the value of the unknown parameters using simple or vanilla **Gradient Decent**.

(a) (**6 pts**) Considering Mean Squared Error (MSE) as the objective function for the quadratic model, find the expression for $\frac{\partial E}{\partial a}$, $\frac{\partial E}{\partial b}$, and $\frac{\partial E}{\partial c}$

(b) (**4 pts**) Assuming the following initial value $a = 0$, $b = 0$, $c = 0$, and $\eta = 0.01$. Perform the first iteration of **Gradient Decent**. Show the working and values of computed gradients as well as the update values of the parameters.

$E = \frac{1}{3}[(2 - (a(1)^2 + b(1) + c))^2 + (6 - (a(2)^2 + b(2) + c))^2 + (12 - (a(3)^2 + b(3) + c))^2]$

$$\frac{\partial E}{\partial a} = -\frac{1}{3}\sum_{i=1}^{4} 2x_i^2(y_i - (ax_i^2 + bx_i + c)) \tag{9}$$

$$\frac{\partial E}{\partial b} = -\frac{1}{3}\sum_{i=1}^{4} 2x_i(y_i - (ax_i^2 + bx_i + c)) \tag{10}$$

$$\frac{\partial E}{\partial c} = -\frac{1}{3}\sum_{i=1}^{4} 2(y_i - (ax_i^2 + bx_i + c)) \tag{11}$$

Iteration 1

- Initial $a = 0, b = 0, c = 0$
- Compute gradients:

$$\frac{\partial MSE}{\partial a} = -\frac{1}{3}\left[2(1^2)(2-(0\cdot 1^2+0\cdot 1+0))+2(2^2)(6-(0\cdot 2^2+0\cdot 2+0))+2(3^2)(12-(0\cdot 3^2+0\cdot 3+0))\right]$$
$$= -\frac{1}{3}[4+48+108] = -\frac{1}{3}\cdot 160 = -53.33$$

$$\frac{\partial MSE}{\partial b} = -\frac{1}{3}\left[2(1)(2-(0\cdot 1^2+0\cdot 1+0))+2(2)(6-(0\cdot 2^2+0\cdot 2+0))+2(3)(12-(0\cdot 3^2+0\cdot 3+0))\right]$$
$$= -\frac{1}{3}[4+24+72] = -\frac{1}{3}\cdot 100 = -33.33$$

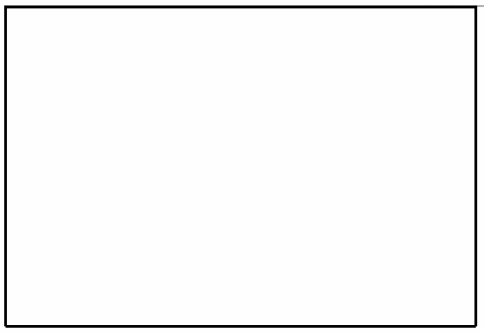
$$\frac{\partial MSE}{\partial c} = -\frac{1}{3}\left[2(2-(0\cdot 1^2+0\cdot 1+0))+2(6-(0\cdot 2^2+0\cdot 2+0))+2(12-(0\cdot 3^2+0\cdot 3+0))\right]$$
$$= -\frac{1}{3}[4+12+24] = -\frac{1}{3}\cdot 40 = -13.33$$

- Update values:

$$a_{\text{new}} = a - \alpha\cdot\frac{\partial MSE}{\partial a} = 0 - 0.01\cdot(-53.33) = 0.533$$
$$b_{\text{new}} = b - \alpha\cdot\frac{\partial MSE}{\partial b} = 0 - 0.01\cdot(-33.33) = 0.333$$
$$c_{\text{new}} = c - \alpha\cdot\frac{\partial MSE}{\partial c} = 0 - 0.01\cdot(-13.33) = 0.133$$

Problem 3. (**3 pts**) Given a 6×6 image, perform a 2×3 max polling.

4	1	4	0	6	7
3	0	3	1	3	0
0	5	0	0	1	0
3	1	3	1	2	0
3	1	0	1	0	3
0	2	0	1	0	1

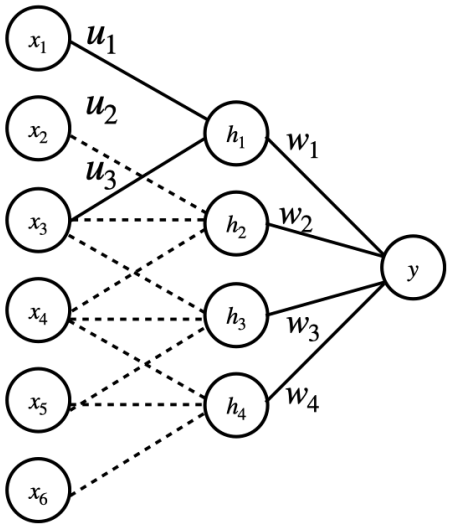


4	6	7
5	3	2
3	1	3

Problem 4. Consider this convolutional neural network architecture.

In the first layer, we have a one-dimensional convolution with a single filter of size 3 such that $h_i = \frac{1}{1+e^{-z}}$, and $z = \sum_{j=1}^{3} u_j x_{i+j-1}$. The second layer is fully connected such that $y = \sum_{i=1}^{4} w_i h_i$. We perform gradient descent on the loss function $\mathcal{J} = (t-y)^2$, where t is the target value or training label for $\mathbf{x}$.

(a) (**1 pts**) What is the total number of parameters in this neural network? Recall that convolutional layers share weights. There are no bias terms.



(b) (**3 pts**) Compute $\frac{\partial \mathcal{J}}{\partial w_i}$

(c) (**4 pts**) Compute $\frac{\partial \mathcal{J}}{\partial u_j}$

Solution

(a) Total 7 parameters (3 in layer 1 and 4 in layer 2)

(b) (**6 pts**) $\frac{\partial \mathcal{J}}{\partial w_i} = -2(t-y)h_i$

(c) (**6 pts**) $\frac{\partial \mathcal{J}}{\partial u_j} = -2(t-y)\frac{\partial y}{\partial u_j} = -2(t-y)\sum_{i=1}^{4}\frac{\partial y}{\partial h_i}\frac{\partial h_i}{\partial u_j} = -2(t-y)\sum_{i=1}^{4} w_i h_i(1-h_i)x_{i+j-1}$

Problem 5. (**5 pts**) Complete the below mentioned table:

Layer	No. of filters	Filter size	stride	padding	output volume	parameters (with bias)
Input	N/A	512x1024	-	-	-	-
Conv	64	3x3	1	1	512x1024x64	(3x3x1+1)64 = 640
Conv	128	5x5	1	0	508x1020x128	(5x5x64+1)128 = 204928
pool	N/A	4x4	4	0	169x340x128	0
Conv	512	3x3	1	0	84x169x512	(3x3x128+1)512 = 590336
FC	N/A	N/A	N/A	N/A	10	(3x3x128+1)512*10 = 72683520

Problem 6. (**6 pts**) Consider a sequence-to-sequence model such as language-to-language translation. The encoder generated the following keys: $k_1 = [2, 3, 1]^T$ $k_2 = [0, 1, 1]^T$ $k_3 = [2, 0, 1]^T$

The decoder generated a query $q_l = [6, 1, 2]^T$. Compute the attention weights.

$e_{1,l} = k_1^T \cdot q_l = [2, 3, 1] \cdot [0.6, 0.1, 0.2]^T = 17$

$e_{2,l} = k_2^T \cdot q_l = [0, 1, 1] \cdot [.6, 0.1, 0.2]^T = 3$

$e_{3,l} = k_3^T \cdot q_l = [2, 0, 1] \cdot [0.6, 0.1, 0.2]^T = 14$

$\sum_t \exp e_{t,l} = \exp e_{1,l} + \exp e_{2,l} + \exp e_{3,l} = 2.4155x10^7 + 20.0855 + 1.2026x10^6 = 2.5358x10^7$

$\alpha_{1,l} = \frac{\exp e_{1,l}}{\sum_t \exp e_{t,l}} = 0.9526$

$\alpha_{2,l} = \frac{\exp e_{1,l}}{\sum_t \exp e_{t,l}} = 7.9209x10^{-7}$

$\alpha_{3,l} = \frac{\exp e_{1,l}}{\sum_t \exp e_{t,l}} = 0.0474$